

Exactly Solvable Quantum Anisotropic Harmonic Oscillator Potentials with Linear and/or Quadratic Perturbations in Higher Dimensions

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Abstract

This paper presents a comprehensive study of rational extensions of the quantum anisotropic and anharmonic oscillator (QAAO) potentials under diverse perturbative frameworks. Beginning with one-dimensional (1D) systems, we demonstrate the existence of rationally extended potentials for imaginary linear perturbations with their exact bound state solutions and derive the corresponding wavefunctions. In two-dimensional (2D) case, we construct the rational extensions for QAAOs with quadratic (λxy) type of perturbations (where λ being real or imaginary) and obtain their solutions analytically. We extend our discussion to the three-dimensional (3D) systems with combinations of linear and quadratic perturbations, as well as combinations of purely quadratic perturbations and obtain their corresponding solutions analytically. For the potentials with imaginary perturbations, we construct the corresponding $\mathcal{PT}$ -symmetric systems, supported by appropriate parity operators, and identify parameter regimes where the spectra remain real (unbroken $\mathcal{PT}$ -symmetry). Additionally, we investigate the constraints on perturbation parameters and angular frequencies required to preserve spectral reality. Various properties of these systems including the conditions for the degeneracy in the higher dimensions are discussed. The behaviors of the potentials with their ground state wavefunctions are also shown graphically.

1 Introduction

The role of quantum harmonic oscillator is well known and has numerous applications across various areas of theoretical as well as experimental physics, particularly in the case involving isotropic or anisotropic materials where the restoring forces are linear in nature. However, for anisotropic materials where the restoring forces are non-linear, such systems are modeled as quantum anisotropic and anharmonic oscillators (QAAO) type. The properties of these anisotropic materials are generally studied by obtaining their exact or approximate solutions. The Quantum mechanics often faces the challenge of finding exact analytical solutions for the Schrödinger equation when dealing with more general or new type of QAAO potentials. With the conjecture that the non-Hermitian Hamiltonians exhibit Parity-Time ($\mathcal{PT}$) symmetry [1] and the subsequent realization that they exhibit real spectra when $\mathcal{PT}$ symmetry remains unbroken [2, 3], the charm of $\mathcal{PT}$ symmetric quantum mechanics has drawn a lot of interest. This has led to the exploration of non-Hermitian quantum systems, particularly those with $\mathcal{PT}$ symmetry, and the search for real spectra in such systems. Subsequently, Mostafazadeh [4, 5, 6] introduced the concept of pseudo-Hermitian Hamiltonians and showed that all the $\mathcal{PT}$ symmetric Hamiltonians studied in the literature exhibited pseudo-Hermiticity. Pseudo-Hermiticity was studied using gauge-like transformation [7], where it was shown that the eigenstates of the $\mathcal{PT}$ -symmetric Hamiltonians fail to satisfy the $\mathcal{PT}$ -symmetric orthogonality conditions, indicating a shortcoming in the theoretical framework of $\mathcal{PT}$ -symmetric quantum mechanics.

Bender and others [1] generalised Bessis's example [8] (which involved the 1D-quantum anharmonic complex potential $x^2 + ix^3$) and considered the potential $x^2 - (ix)^N$ for real N . They demonstrated that the spectrum is real at various $\mathcal{PT}$ phase transition points. In the work by Fernández

[9], a general coordinate-momentum transformation was shown, which simplified the derivation of Ahmed [7] and Rath and Mallick [10]. In the available literature, the solution of one dimensional harmonic oscillator ($\omega_1^2 x^2$) with a linear type perturbation ($\lambda_0 x$) is well known [11] and solved simply by coordinate shifting. The complex and $\mathcal{PT}$ symmetric version of this potential is equivalent to one of the simplest form of the Bender's problem for $N = 1$, which is exactly solvable and the spectrum is always real for any values of λ . The same is further extended to the two-dimensional hermitian QAAO case with quadratic type perturbation ($\frac{\lambda_1}{2} xy$) and solved the system analytically. The system was made non-hermitian but $\mathcal{PT}$ symmetric by considering λ as imaginary and shown that the system will exhibit real spectra under unbroken $\mathcal{PT}$ symmetry [12]. Recently, after the discovery of exceptional orthogonal polynomials (EOPs) [13, 14, 15], it has been observed that many conventional potentials have been rationally extended and obtained their bound as well as scattering states solutions using the expressions of these EOPs [16]. Similarly, the conventional 1D harmonic oscillator potential is also extended and obtained their solutions in terms of exceptional Hermite polynomials for even co-dimensions of m [12] and discussed its different properties in details in ref. [17]. This problem is further extended for the case of higher dimensional quantum anisotropic harmonic oscillator (QAHO) potentials for both even as well as odd co-dimensions of m and obtained their solutions in terms of exceptional Hermite and the exceptional Laguerre polynomials [18, 19]. As far our knowlwdge is concerned the exact solution of a more general type of harmonic oscillator potentials in three dimension with combination of quadratic and linear perturbations ($\lambda xy + \lambda_o z$) or combination of quadratic perturbation ($\lambda_1 xy + \lambda_2 yz + \lambda_3 xz$) with λ 's being real or imaginary are not available in the literature and needs more attention. Further, it will be more interesting to construct the rationally extended real as well as complex and $\mathcal{PT}$ symmetric potentials corresponding to these conventional QAAO potentials with their various other properties.

This manuscript addresses the problem by deriving analytical solutions for potentials with quadratic pururbations in higher dimensions through coordinate transformation. We begin by considering the potential, which includes either linear or quadratic perturbations. In the first step, we eliminate the perturbation terms by transforming to the tilde coordinates, resulting in a pure anisotropic potential. Transformations include coordinate shifting to absorb the linear perturbation and coordinate rotation to absorb the quadratic perturbation. In the second step, we extend this potential rationally, following the well-known procedure discussed in Refs. [20] and [16]. The resulting rationally extended (RE) potential in the tilde coordinates is parameterized by m . Finally, applying the inverse coordinate transformation, we obtain the rational extension of the original perturbed potential. In all these transformations, Laplacian operators ∇^2 remains invariant. We also give an expression for the transformation realtion between new and old coordinates and show that $\mathcal{PT}$ -symmetric Hamiltonians are obtained for selected parity operators.

The structure of our presentation is organized as follows: In sections 2, we delve into a thorough analysis of real and $\mathcal{PT}$ -symmetric complex perturbed quantum harmonic oscillators (QHO) in 1D and generate a corresponding new family of m -dependent RE potentials. We show that the RE real potential is acceptable only for even m . However, the complex and PT symmetric case is defined for all integers (even and odd) of m . In section 3, we consider 2D real and complex $\mathcal{PT}$ -symmetric QAAO potential with quadratic perturbations defined on the full-line and discuss the various conditions for the spectrum to be real. These potentials are then extended rationally for even co-dimensions of m_1 and m_2 , and their solutions are obtained in terms of exceptional Hermite polynomials. The degeneracy conditions, along with some examples in both real and $\mathcal{PT}$ -symmetric cases, are also discussed. Thereafter, a 3D case of QAAO potentials is considered with combinations of linear and/or quadratic perturbations and constructed a family of anisotropic 3D QAHO potentials with their analytical solutions in section 4. We show that the pseudo-Hermitian Hamiltonians are $\mathcal{PT}$ -symmetric for selected parity operators, and we obtain their exact solutions and finally construct the corresponding RE potentials with perturbation terms. Finally in Section 5, we sumarize our findings.

2 Construction of one-dimensional rationally extended potentials

Let's start with a well-known one-dimensional quantum harmonic oscillator (QHO) potential $V(\omega_x, x)$ (defined on the full line i.e., $-\infty < x < \infty$) with ω_1 as the angular frequency satisfying the Schrödinger equation in units of $\hbar = 2M = 1$, given by

$$\left[-\frac{d^2}{dx^2} + V(x) \right] \Psi = E\Psi \quad \text{where} \quad V(x) = \frac{1}{4}\omega_1^2 x^2. \quad (1)$$

we now introduce a linear perturbation $\lambda_0 x$ given by

$$V^l(x) = V(x) + \lambda_0 x, \quad (2)$$

which can be reduced to the standard form of QHO [21] by shifting the coordinate $x \rightarrow \tilde{x}$ and get

$$\tilde{V}^l(\tilde{x}) = \frac{1}{4}\omega_1^2 \tilde{x}^2 - \frac{\lambda_0^2}{\omega_1^2} \quad \text{where} \quad \tilde{x} = x + \frac{2\lambda_0}{\omega_1^2}, \quad (3)$$

The tilde quantity in rest sections of this paper represents the expression in new coordinate. The solution of this potential is same as the QHO except the energy eigenvalues are shifted by λ_0^2/ω_1^2 . The spectra of Eq. (3) are real for any value of λ_0 (real or imaginary) and for imaginary λ_0 i.e., $\lambda_0 = i\gamma_0$, the system is complex and $\mathcal{PT}$ symmetric. The energy eigenvalues

$$E \rightarrow \tilde{E}_n^l(\omega_1) = \left(n + \frac{1}{2} \right) \omega_1 + \frac{\gamma_0^2}{\omega_1^2}; \quad n = 0, 1, 2, \dots \quad (4)$$

The wavefunctions are the same as those of the QHO except for a shift in the coordinate $x \rightarrow \tilde{x}$. Therefore, the wavefunctions are given by

$$\tilde{\psi}_n^l(\tilde{x}) = \psi_n(\tilde{x}), \quad \text{Here,} \quad \tilde{\omega}_1 = \omega_1. \quad (5)$$

which exhibit unbroken $\mathcal{PT}$ symmetry for any real values of γ_0 and satisfy

$$\mathcal{PT} \tilde{\psi}_n^l(\tilde{x}) = (-1)^n \tilde{\psi}_n^l(\tilde{x}) \quad (6)$$

2.1 Rational extension

The rational extension of the potential (1) defined on the full line can be easily obtained by using the results discussed in [20, 16] for even co-dimension¹ of m and we list potential, wavefunction and energy of rationally extended Quantum Harmonic Oscillator (RE-QHO) potential as the results will be used in the coming sections.

$$V_{RE,m}(x) = V(x) + V_{rat,m}(x), \quad -\infty < x < \infty \quad (7)$$

where $V_{rat,m}(x)$ is given by

$$V_{rat,m}(x) = -2 \left[\frac{\mathcal{H}_m''(\sqrt{\frac{\omega_1}{2}}x)}{\mathcal{H}_m(\sqrt{\frac{\omega_1}{2}}x)} - \left[\frac{\mathcal{H}_m'(\sqrt{\frac{\omega_1}{2}}x)}{\mathcal{H}_m(\sqrt{\frac{\omega_1}{2}}x)} \right]^2 + \frac{\omega_1}{2} \right] \quad (8)$$

¹If we consider the potential defined on the half-line, the corresponding rational extension will be acceptable for all positive integer values of m [16]. In this manuscript, we consider the potential defined on the full-line only. The same procedure may also be adopted to handle the potentials defined on the half-line.

with $m = 0, 2, 4, \dots$. The ground state, excited state and energy eigen values are given as

$$\begin{aligned}\psi_{RE,m,0}(x) &= \frac{\zeta(\omega_1, x)}{\mathcal{H}_m\left(\sqrt{\frac{\omega_1}{2}}x\right)} \\ \text{and } \psi_{RE,m,n+1}(x) &= \frac{\zeta(\omega_1, x)}{\mathcal{H}_m\left(\sqrt{\frac{\omega_1}{2}}x\right)} \hat{H}_{m,n+1}\left(\sqrt{\frac{\omega_1}{2}}x\right),\end{aligned}\quad (9)$$

with $\zeta(\omega_1, x) = e^{-\frac{\omega_1}{4}x^2}$ respectively, where $n = 0, 1, 2, \dots$ and $\hat{H}_{m,0}(x) = 1$. Here $\hat{H}_{m,n+1}(x)$ is the exceptional Hermite polynomial [20] given by

$$\hat{H}_{m,n+1}(x) = \mathcal{H}_m(x)H_{n+1}(x) + H_n(x)\frac{d}{dx}\mathcal{H}_m(x), \quad (10)$$

where $H_n(x)$ is the classical Hermite polynomial and $\mathcal{H}_m(x)$ is the pseudo Hermite polynomial. The energy eigenvalues [16] are given by

$$E_{RE,m,n+1}(\omega_1) = (n + m + 1)\omega_1 \quad \text{with} \quad E_{RE,m,0}(\omega_1) = 0. \quad (11)$$

Therefore the rational extension of linearly perturbed potential (3) in new coordinate $\tilde{V}^l(\tilde{x})$ is obtained using above results as

$$\tilde{V}_{RE,m}^l(\tilde{x}) = \tilde{V}^l(\tilde{x}) + V_{rat,m}(\tilde{x}), \quad -\infty < x < \infty \quad (12)$$

with new frequency $\tilde{\omega}_1 = \omega_1$. In the old coordinates $\tilde{V}_{RE,m}^l(\tilde{x})$ changes to $V_{RE,m}^l(x)$ and we have perturbation term. When λ_0 is real, $\tilde{x}$ is also real, and the potential becomes singular for odd m due to the presence of a $\frac{1}{\tilde{x}^2}$ term, which diverges at the origin. As a result, only even m values are allowed. While even m may also appear to diverge at certain values of $\tilde{x}$, the polynomial in the denominators of the rational terms has only complex roots, preventing such divergences. The corresponding ground and the excited state eigenfunctions are given by

$$\begin{aligned}\tilde{\psi}_{RE,m,0}^l(\tilde{x}) &= \frac{\zeta(\omega_1, \tilde{x})}{\mathcal{H}_m\left(\sqrt{\frac{\omega_1}{2}}\tilde{x}\right)} \\ \text{and } \tilde{\psi}_{RE,m,n+1}^l(\tilde{x}) &= \frac{\zeta(\omega_1, \tilde{x})}{\mathcal{H}_m\left(\sqrt{\frac{\omega_1}{2}}\tilde{x}\right)} \hat{H}_{m,n+1}\left(\sqrt{\frac{\omega_1}{2}}\tilde{x}\right).\end{aligned}\quad (13)$$

The energy eigenvalues of $\tilde{V}_{RE,m}^l(\tilde{x})$ are same as $V_{RE,m}(x)$ as energy factor of $\frac{\lambda_0^2}{\omega_1^2}$ cancels out [16] and are given by

$$\tilde{E}_{RE,m,n+1}^l(\omega_1) = (n + m + 1)\omega_1 \quad \text{with} \quad \tilde{E}_{RE,m,0}^l(\omega_1) = 0. \quad (14)$$

2.1.1 Complex and $\mathcal{PT}$ symmetric case

For imaginary $\lambda_0 = i\gamma_0$, the extended potential (12) is complex, $\mathcal{PT}$ -symmetric, and well-defined for all m , as can be verified from Table 1, which presents it in terms of λ_0 and in the old coordinates. Similar to the real case, in this case the energy eigenvalues are also real and independent of the parameter γ_0 . The ground and the excited state wavefunctions (13) also satisfy

$$\begin{aligned}\mathcal{PT} \tilde{\psi}_{RE,m,0}^l(\tilde{x}) &= \tilde{\psi}_{RE,m,0}^l(\tilde{x}) \\ \text{and } \mathcal{PT} \tilde{\psi}_{RE,m,n+1}^l(\tilde{x}) &= (-1)^{n+1} \tilde{\psi}_{RE,m,n+1}^l(\tilde{x})\end{aligned}\quad (15)$$

respectively. The potential in Eq. (13) when expressed in terms of the old coordinates x has the following form as listed in Table 1. The corresponding ground and the excited state eigenfunctions for the first few values of m are listed in Table 2 and Table 3, respectively. The plot for the imaginary value of λ_0 , specifically $\lambda_0 = i$, is provided in Fig. 1. The corresponding ground-state wavefunction is shown in Fig. 2.

m	$V_{\text{RE},m}^l(\mathbf{x})$
0	$\frac{\omega_1^2 x^2}{4} + \lambda_0 x - \omega_1$
1	$\frac{\omega_1^2 x^2}{4} + \lambda_0 x - \omega_1 + \frac{2\omega_1^4}{(2\lambda_0 + \omega_1^2 x)^2}$
2	$\frac{\omega_1^2 x^2}{4} + \lambda_0 x - \omega_1 - \frac{8\omega_1^7}{(4\lambda_0^2 + \omega_1^3 + \omega_1^4 x^2 + 4\lambda_0 \omega_1^2 x)^2} + \frac{4\omega_1^4}{4\lambda_0^2 + \omega_1^3 + \omega_1^4 x^2 + 4\lambda_0 \omega_1^2 x}$
3	$\frac{\omega_1^2 x^2}{4} + \lambda_0 x - \omega_1 - \frac{24\omega_1^7}{(4\lambda_0^2 + 3\omega_1^3 + \omega_1^4 x^2 + 4\lambda_0 \omega_1^2 x)^2} + \frac{4\omega_1^4}{4\lambda_0^2 + 3\omega_1^3 + \omega_1^4 x^2 + 4\lambda_0 \omega_1^2 x} + \frac{2\omega_1^4}{(2\lambda_0 + \omega_1^2 x)^2}$

Table 1: Potentials $V_{RE,m}^l(x)$ for different m values in old coordinates.

m	$\psi_{\text{RE},m,0}^l(\mathbf{x})$
0	$\zeta\left(\omega_1, x + \frac{2\lambda_0}{\omega_1^2}\right)$
1	$\zeta\left(\omega_1, x + \frac{2\lambda_0}{\omega_1^2}\right) \left[\frac{(-\omega_1)^{3/2}}{\sqrt{2}(2\lambda_0 + x\omega_1^2)} \right]$
2	$\zeta\left(\omega_1, x + \frac{2\lambda_0}{\omega_1^2}\right) \left[-\frac{\omega_1^3}{2(4\lambda_0^2 + 4x\lambda_0\omega_1^2 + \omega_1^3 + x^2\omega_1^4)} \right]$
3	$\zeta\left(\omega_1, x + \frac{2\lambda_0}{\omega_1^2}\right) \left[\frac{(-\omega_1)^{9/2}}{2\sqrt{2}(2\lambda_0 + x\omega_1^2)(4\lambda_0^2 + 4x\lambda_0\omega_1^2 + \omega_1^3(3 + x^2\omega_1))} \right]$

Table 2: Table for $\psi_{RE,m,0}^l(x)$ with m values 0 to 3 in old coordinates.

m	$\psi_{\text{RE},m,n+1}^l(\mathbf{x})$
0	$\zeta\left(\omega_1, x + \frac{2\lambda_0}{\omega_1^2}\right) \left[\frac{(2\lambda_0 + \omega_1^2 x)}{\omega_1} H_n\left(\sqrt{\frac{\omega_1}{2}}\left(x + \frac{2\lambda_0}{\omega_1^2}\right)\right) - \sqrt{2}n\sqrt{\omega_1}H_{n-1}\left(\sqrt{\frac{\omega_1}{2}}\left(x + \frac{2\lambda_0}{\omega_1^2}\right)\right) \right]$
1	$\zeta\left(\omega_1, x + \frac{2\lambda_0}{\omega_1^2}\right) \left[\frac{(4\lambda_0^2 + \omega_1^3 + \omega_1^4 x^2 + 4\lambda_0 \omega_1^2 x)}{2\lambda_0 \omega_1 + \omega_1^3 x} H_n\left(\sqrt{\frac{\omega_1}{2}}\left(x + \frac{2\lambda_0}{\omega_1^2}\right)\right) - \sqrt{2}n\omega_1^{3/2} \frac{(2\lambda_0 + \omega_1^2 x)}{\omega_1} H_{n-1}\left(\sqrt{\frac{\omega_1}{2}}\left(x + \frac{2\lambda_0}{\omega_1^2}\right)\right) \right]$
2	$\zeta\left(\omega_1, x + \frac{2\lambda_0}{\omega_1^2}\right) \left[\frac{(8\lambda_0^3 + 6\lambda_0\omega_1^3(\omega_1 x^2 + 1) + \omega_1^5 x(\omega_1 x^2 + 3) + 12(\lambda_0^2)\omega_1^2 x)}{4\lambda_0^2 \omega_1 + \omega_1^4 + \omega_1^5 x^2 + 4\lambda_0 \omega_1^3 x} H_n\left(\sqrt{\frac{\omega_1}{2}}\left(x + \frac{2\lambda_0}{\omega_1^2}\right)\right) - \sqrt{2}n\sqrt{\omega_1}H_{n-1}\left(\sqrt{\frac{\omega_1}{2}}\left(x + \frac{2\lambda_0}{\omega_1^2}\right)\right) \right]$
3	$\zeta\left(\omega_1, x + \frac{2\lambda_0}{\omega_1^2}\right) \frac{(16\lambda_0^4 + 24(\lambda_0^2)\omega_1^3(\omega_1 x^2 + 1) + 8\lambda_0\omega_1^5 x(\omega_1 x^2 + 3) + \omega_1^6(\omega_1^2 x^4 + 6\omega_1 x^2 + 3) + 32\lambda_0^3\omega_1^2 x)}{\omega_1(2\lambda_0 + \omega_1^2 x)(4\lambda_0^2 + \omega_1^3(\omega_1 x^2 + 3) + 4\lambda_0 \omega_1^2 x)} H_n\left(\sqrt{\frac{\omega_1}{2}}\left(x + \frac{2\lambda_0}{\omega_1^2}\right)\right)$

Table 3: Table for $\psi_{RE,m,n+1}^l(x)$ with m values 0 to 3 in old coordinates.

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Figure 1: Plots of Real and Imaginary components of $V_{RE,m}^l(x)$ vs x for the first few m with $\omega_x = 2$.

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Figure 2: Plots of Real and Imaginary components of $\psi_{RE,m,0}^l(x)$ vs x for the first few m with $\omega_x = 2$.

3 Two-dimensional QAAO potential with quadratic perturbation

In this section, we consider a 2D quantum anisotropic and anharmonic oscillator potential with a quadratic perturbation term $\frac{\lambda}{2}xy$ (with λ being real or imaginary) and angular frequencies ω_1 and ω_2 along the x and y axes, respectively, given by

$$V(x, y) = \frac{1}{4}\omega_1^2 x^2 + \frac{1}{4}\omega_2^2 y^2 + \frac{\lambda}{2}xy. \quad (16)$$

The transformation coordinates [12] from old to new coordinates and vice versa are given by

$$\begin{cases} x = a\tilde{x} + b\tilde{y}, \\ y = -b\tilde{x} + a\tilde{y}, \end{cases} \quad \begin{cases} \tilde{x} = ax - by, \\ \tilde{y} = bx + ay, \end{cases} \quad (17)$$

where $k = \frac{\omega_1^2 - \omega_2^2}{\sqrt{4\lambda^2 + (\omega_1^2 - \omega_2^2)^2}}$, and $a = \sqrt{\frac{1-k}{2}}$, $b = \sqrt{\frac{1+k}{2}}$. The angular frequencies in the new coordinates ($\tilde{\omega}_1$ and $\tilde{\omega}_2$) are given by

$$\begin{aligned} \tilde{\omega}_1 &= \sqrt{\frac{1}{2} \left(\omega_1^2 + \omega_2^2 - \sqrt{4\lambda^2 + (\omega_1^2 - \omega_2^2)^2} \right)} \\ \text{and } \tilde{\omega}_2 &= \sqrt{\frac{1}{2} \left(\omega_1^2 + \omega_2^2 + \sqrt{4\lambda^2 + (\omega_1^2 - \omega_2^2)^2} \right)} \end{aligned} \quad (18)$$

respectively. On using the above co-ordinate transformations (17), the resulting uncoupled potential in the new co-ordinates, $\tilde{V}(\tilde{x}, \tilde{y})$ becomes

$$\tilde{V}(\tilde{x}, \tilde{y}) = \tilde{V}(\tilde{x}) + \tilde{V}(\tilde{y}). \quad (19)$$

where, $\tilde{V}(\tilde{x})$, $\tilde{V}(\tilde{y})$ represents QHO potential (1) in new coordinates with new angular frequencies. Similarly corresponding wavefunctions $\tilde{\psi}_{n_1, n_2}(\tilde{x}, \tilde{y})$ and the energy eigenvalues $\tilde{E}_{n_1, n_2}(\tilde{\omega}_1, \tilde{\omega}_2)$ are given by

$$\tilde{\psi}_{n_1, n_2}(\tilde{x}, \tilde{y}) \propto \tilde{\psi}_{n_1}(\tilde{x}) \tilde{\psi}_{n_2}(\tilde{y}), \quad n_1, n_2 = 0, 1, 2, \dots, \quad (20)$$

$$\text{and } \tilde{E}_{n_1, n_2}(\tilde{\omega}_1, \tilde{\omega}_2) = \left(n_1 + \frac{1}{2} \right) \tilde{\omega}_1 + \left(n_2 + \frac{1}{2} \right) \tilde{\omega}_2, \quad (21)$$

respectively, where $\tilde{\psi}_n(\tilde{x})$ is the eigenfunction of the 1D-QHO in new coordinates with new angular frequencies defined by (18). The spectrum is real within specific ranges when λ is either real or purely imaginary, discussed below as the hermitian and the non-hermitian $\mathcal{PT}$ symmetric cases, respectively.

Case(a) Hermitian case: When λ is real, the conditions for real spectra is

$$\sqrt{4\lambda^2 + (\omega_1^2 - \omega_2^2)^2} \leq \omega_1^2 + \omega_2^2, \quad (22)$$

which implies a restriction on λ i.e., $|\lambda| \leq \omega_1\omega_2$. The potential $V(x, y)$ satisfy η -pseudo hermicity condition as η is an identity $\mathbb{I}$ for this case.

Case(b) Non-hermitian and $\mathcal{PT}$ symmetric case: When λ is imaginary, say $\lambda = i\gamma$, the system is non-hermitian and $\mathcal{PT}$ symmetric and the spectrum will be real only when the following conditions must be satisfied i.e.,

$$\begin{aligned} &\sqrt{-4\gamma^2 + (\omega_1^2 - \omega_2^2)^2} \leq \omega_1^2 + \omega_2^2, \\ \text{and } &-4\gamma^2 + (\omega_1^2 - \omega_2^2)^2 > 0. \end{aligned} \quad (23)$$

These conditions imply a restriction on λ i.e., $|\lambda| < \frac{1}{2}|\omega_1^2 - \omega_2^2|$. The η for imaginary λ is not identity and is given by

$$\eta = \begin{pmatrix} -k & -\sqrt{1-k^2} \\ \sqrt{1-k^2} & -k \end{pmatrix} \quad (24)$$

and the potential $V(x, y)$ satisfy η -pseudo hermicity condition as

$$V^\dagger = \eta V \eta^{-1}. \quad (25)$$

The system is $\mathcal{PT}$ symmetric for the parity operators P_1 and P_2 given in (B.3) and the eigenfunction, $\tilde{\psi}_{n_1, n_2}(\tilde{x}, \tilde{y})$, satisfies the following relations [22]

$$\begin{aligned} P_1 T \tilde{\psi}_{n_1, n_2}(\tilde{x}, \tilde{y}) &= (-1)^{n_1} \tilde{\psi}_{n_1, n_2}(\tilde{x}, \tilde{y}) = \pm \tilde{\psi}_{n_1, n_2}(\tilde{x}, \tilde{y}), \\ P_2 T \tilde{\psi}_{n_1, n_2}(\tilde{x}, \tilde{y}) &= (-1)^{n_2} \tilde{\psi}_{n_1, n_2}(\tilde{x}, \tilde{y}) = \pm \tilde{\psi}_{n_1, n_2}(\tilde{x}, \tilde{y}), \end{aligned}$$

and hence the $\mathcal{PT}$ symmetry is unbroken.

3.1 Rational Extension

At the outset, it is important to note that the rational extension of the quantum anharmonic oscillator (QAO) potentials in two dimensions can be performed in four distinct ways [16], leading to four different forms of rationally extended potentials. These possibilities include both x and y on the full line, both on the half line, and one on the full line while the other is on the half line. For the sake of simplicity, we consider one specific case (when both x and y defined on full axes) in this manuscript. The other three cases follow directly from the same approach and can be easily derived.

As shown in our recent work [16] that the rational extension of these QAO potentials in two or higher dimensions is simply the sum of the rational extensions of QAO potentials along each dimension. Here again, only even codimensions are possible, as the new coordinates $\tilde{x}$ and $\tilde{y}$ are linear functions of the old coordinates x and y . Consequently, when the old coordinates approach zero, the new coordinates also approach zero, causing the $\frac{1}{x^2}$ and $\frac{1}{y^2}$ terms, which appear for odd m , to diverge. Thus, the rationally extended potential corresponding to the potential (19) for even co-dimensions of m_1 and m_2 will be

$$\tilde{V}_{RE, m_1, m_2}(\tilde{x}, \tilde{y}) = \tilde{V}_{RE, m_1}(\tilde{x}) + \tilde{V}_{RE, m_2}(\tilde{y}), \quad (26)$$

where $\tilde{V}_{RE, m_1}(\tilde{x})$ and $\tilde{V}_{RE, m_2}(\tilde{y})$ are the rational extension of QHO in new coordinates along $\tilde{x}$ and $\tilde{y}$ axes. Doing inverse coordinate transformation we get the expression of rationally extended potential $V_{RE, m_1, m_2}(x, y)$ in old coordinates but in terms of new frequencies $\tilde{\omega}_1$ and $\tilde{\omega}_2$ as shown in Table-4 for the first few values of m_1 and m_2 . The corresponding eigenfunctions and the energy eigenvalues are given using (9) and (11) in new coordinates as

$$\tilde{\psi}_{RE, m_1, m_2, 0, 0}(\tilde{x}, \tilde{y}) \propto \tilde{\psi}_{RE, m_1, 0}(\tilde{x}) \tilde{\psi}_{RE, m_2, 0}(\tilde{y}) \quad (27)$$

$$\tilde{\psi}_{RE, m_1, m_2, n_1+1, n_2+1}(x, y) \propto \tilde{\psi}_{RE, m_1, n_1+1}(\tilde{x}) \tilde{\psi}_{RE, m_2, n_2+1}(\tilde{y}) \quad (28)$$

$$\text{and } \tilde{E}_{RE, m_1, m_2, n_1+1, n_2+1}(\tilde{\omega}_1, \tilde{\omega}_2) = \tilde{E}_{RE, m_1, n_1+1}(\tilde{\omega}_1) + \tilde{E}_{RE, m_2, n_2+1}(\tilde{\omega}_2) \quad (29)$$

$$\text{with } \tilde{E}_{RE, m_1, m_2, 0, 0}(\tilde{\omega}_1, \tilde{\omega}_2) = 0, \quad n_1, n_2 = 0, 1, 2, \dots,$$

respectively. The rationally extended potential $\tilde{V}_{RE, m_1, m_2}(\tilde{x}, \tilde{y})$ is regular for even m irrespective of λ being real or imaginary. In the real case, the spectrum will be real only when the same condition (22) as defined in the conventional case is satisfied. However, in the case of imaginary $\lambda = i\gamma$, the potential $\tilde{V}_{RE, m_1, m_2}(\tilde{x}, \tilde{y})$ is also $\mathcal{PT}$ symmetry under both the parity operators $P \rightarrow P_1$ or P_2 (B.3)

and the spectrum is unbroken under the same restrictions defined for the conventional case (23). The ground and the excited state eigenfunctions corresponding to this RE potentials also satisfy

$$\begin{aligned} PT \tilde{\psi}_{RE,m_1,m_2,0,0}(\tilde{x}, \tilde{y}) &= \tilde{\psi}_{RE,m_1,m_2,0,0}(\tilde{x}, \tilde{y}) \\ PT \tilde{\psi}_{m_1,m_2,n_1+1,n_2+1}(\tilde{x}, \tilde{y}) &= (-1)^{n_1+n_2+2} \tilde{\psi}_{m_1,m_2,n_1,n_2}(\tilde{x}, \tilde{y}) = \pm \tilde{\psi}_{m_1,m_2,n_1,n_2}(\tilde{x}, \tilde{y}). \end{aligned}$$

respectively.

3.2 Conditions for Degeneracy

The system will show degeneracy when the ratio of angular frequencies in new coordinates is a rational number. i.e.,

$$\frac{\tilde{\omega}_1}{\tilde{\omega}_2} = \frac{\tilde{r}}{\tilde{s}}, \quad (30)$$

in the potential $\tilde{V}(\tilde{x}, \tilde{y})$ given by

$$\tilde{V}(\tilde{x}, \tilde{y}) = \frac{1}{4} (\tilde{\omega}_1^2 \tilde{x}^2 + \tilde{\omega}_2^2 \tilde{y}^2). \quad (31)$$

Similarly $\frac{r}{s}$ represents the frequency ratio $\frac{\omega_1}{\omega_2}$ in old coordinates which can be irrational. Substituting the values of $\tilde{\omega}_1$ and $\tilde{\omega}_2$ from Eq. (18) and simplifying for λ (depends on $\tilde{r}, \tilde{s}, r, s, \omega_1, \omega_2$) we get

$$\lambda(r, s, \tilde{r}, \tilde{s}, \omega_1, \omega_2) = \frac{\sqrt{(\tilde{r}^4 + \tilde{s}^4) - \left(\frac{\tilde{r}\tilde{s}}{rs}\right)^2 (r^4 + s^4)}}{\tilde{r}^2 + \tilde{s}^2} \omega_1 \omega_2 \quad (32)$$

The energy eigenvalues of degenerate potential (31) are given by

$$\tilde{E}_{n_1,n_2}(\tilde{r}, \tilde{s}, \tilde{\omega}_2) = \left(\tilde{r}n_1 + \tilde{s}n_2 + \frac{\tilde{r} + \tilde{s}}{2} \right) \frac{\tilde{\omega}_2}{\tilde{s}}. \quad (33)$$

It can be seen that the degeneracy is seen only when λ is satisfied by the above equation. To illustrate, we consider some explicit examples of real and $\mathcal{PT}$ symmetric cases discussed below:

3.2.1 Examples of real case

Example 1: For equal angular frequencies $\omega_1 = \omega_2 = \omega$

This is the case of an isotropic oscillator with perturbation. In this case, the parameter a and b in (17) is $\sqrt{\frac{1}{2}}$ as $k = 0$ and therefore the coordinate transformation relation from old to new coordinates and new angular frequencies simplifies to

$$\begin{aligned} \tilde{x} &= \frac{1}{\sqrt{2}} (x + y), \\ \tilde{y} &= \frac{1}{\sqrt{2}} (y - x), \end{aligned} \quad \begin{cases} \tilde{\omega}_1 = \sqrt{\omega^2 - \lambda}, \\ \tilde{\omega}_2 = \sqrt{\omega^2 + \lambda}. \end{cases} \quad (34)$$

The degeneracy condition given by Eq. (32) simplifies to

$$\lambda(\tilde{r}, \tilde{s}, \omega) = \frac{|\tilde{r}^2 - \tilde{s}^2|}{\tilde{r}^2 + \tilde{s}^2} \omega^2. \quad (35)$$

When λ is zero then $\tilde{r} = \tilde{s}$ as a result energy eigenvalues given by Eq. 33 becomes

$$E_{\lambda=0} = (n_1 + n_2 + 1) \omega \quad (36)$$

and the system is $n + 1$ -fold degenerate, where $n = n_1 + n_2$ and represents the n^{th} excited state.

Example 2: Irrational Angular Frequencies $(\omega_1, \omega_2, \lambda) = (1, \sqrt{2}, \frac{\sqrt{14}}{13})$

The potential in Eq. (16) can be reduced to a standard form of QHO using the transformation given by Eq. (17) with $k = -\frac{13}{\sqrt{15}}$ giving

$$\tilde{V}(\tilde{x}, \tilde{y}) = \frac{1}{4} \left(\frac{12}{13} \tilde{x}^2 + \frac{27}{13} \tilde{y}^2 \right)$$

The angular frequencies in new coordinates have ratio $\tilde{\omega}_1 : \tilde{\omega}_2 = 2 : 3$ and the system is degenerate.

Example 3: Rational Angular Frequencies $(\omega_1, \omega_2, \lambda) = (1, 2, \frac{\sqrt{7}}{2})$

The perturbed potential has the form

$$V(x, y) = \frac{1}{4} \left(\frac{1}{2} x^2 + \frac{9}{2} y^2 \right) + \frac{\sqrt{7}}{4} xy. \quad (37)$$

Doing coordinate transformation similar to the above example, but using $k = -\frac{3}{4}$ gives

$$\tilde{V}(\tilde{x}, \tilde{y}) = \frac{1}{4} \left(\frac{1}{2} \tilde{x}^2 + \frac{9}{2} \tilde{y}^2 \right),$$

and the angular frequencies in new coordinates have ratio $\tilde{\omega}_1 : \tilde{\omega}_2 = 1 : 3$ and the system is degenerate. The corresponding rational extension $V_{RE,0,2}(x, y)$ for $m_1, m_2 = 0, 2$ and $m_1, m_2 = 2, 2$ are

$$V_{RE,0,2}(x, y) = V(x, y) - \frac{1}{\sqrt{2}} - 2 \left(\frac{3}{2\sqrt{2}} - \frac{72 \left(\frac{x}{2\sqrt{2}} + \frac{1}{2} \sqrt{\frac{7}{2}} y \right)^2}{\left(-2 - 3\sqrt{2} \left(\frac{x}{2\sqrt{2}} + \frac{1}{2} \sqrt{\frac{7}{2}} y \right)^2 \right)^2} - \frac{6\sqrt{2}}{-2 - 3\sqrt{2} \left(\frac{x}{2\sqrt{2}} + \frac{1}{2} \sqrt{\frac{7}{2}} y \right)^2} \right)$$

and

$$V_{RE,2,2}(x, y) = V(x, y) - 2 \left(\frac{1}{2\sqrt{2}} - \frac{8 \left(\frac{1}{2} \sqrt{\frac{7}{2}} x - \frac{y}{2\sqrt{2}} \right)^2}{\left(-2 - \sqrt{2} \left(\frac{1}{2} \sqrt{\frac{7}{2}} x - \frac{y}{2\sqrt{2}} \right)^2 \right)^2} - \frac{2\sqrt{2}}{-2 - \sqrt{2} \left(\frac{1}{2} \sqrt{\frac{7}{2}} x - \frac{y}{2\sqrt{2}} \right)^2} \right) - 2 \left(\frac{3}{2\sqrt{2}} - \frac{72 \left(\frac{x}{2\sqrt{2}} + \frac{1}{2} \sqrt{\frac{7}{2}} y \right)^2}{\left(-2 - 3\sqrt{2} \left(\frac{x}{2\sqrt{2}} + \frac{1}{2} \sqrt{\frac{7}{2}} y \right)^2 \right)^2} - \frac{6\sqrt{2}}{-2 - 3\sqrt{2} \left(\frac{x}{2\sqrt{2}} + \frac{1}{2} \sqrt{\frac{7}{2}} y \right)^2} \right)$$

respectively. The plot of the potential is shown in Fig. 3.

3.2.2 Example of complex and $\mathcal{PT}$ symmetric case

As in the real case, the system is also degenerate in the complex and $\mathcal{PT}$ -symmetric cases. For example, if we consider $(\omega_1, \omega_2, \lambda) = (1, 3, i\sqrt{7})$, the conventional potential has the form

$$V(x, y) = \frac{1}{4} (x^2 + 9y^2) + i \frac{\sqrt{7}}{2} xy. \quad (38)$$

Using the transformation given by Eq. (17) with $k = -\frac{4}{3}$ reduces the anharmonic potential to

$$\tilde{V}(\tilde{x}, \tilde{y}) = \frac{1}{4} (2\tilde{x}^2 + 8\tilde{y}^2)$$

and therefore the ratio of angular frequencies in new coordinates is $\tilde{\omega}_1 : \tilde{\omega}_2 = 1 : 2$ and the system is degenerate. The rationally extended potential corresponding to the potential (38) for $m_1, m_2 = 2, 2$ will be

$$V_{RE,2,2}(x, y) = V(x, y) - 2 \left(-\frac{128 \left(\sqrt{\frac{7}{6}}y + \frac{ix}{\sqrt{6}} \right)^2}{\left(-2 - 4\sqrt{2} \left(\sqrt{\frac{7}{6}}y + \frac{ix}{\sqrt{6}} \right)^2 \right)^2} - \frac{8\sqrt{2}}{-2 - 4\sqrt{2} \left(\sqrt{\frac{7}{6}}y + \frac{ix}{\sqrt{6}} \right)^2} + \sqrt{2} \right) \\ - 2 \left(-\frac{32 \left(\sqrt{\frac{7}{6}}x - \frac{iy}{\sqrt{6}} \right)^2}{\left(-2 - 2\sqrt{2} \left(\sqrt{\frac{7}{6}}x - \frac{iy}{\sqrt{6}} \right)^2 \right)^2} - \frac{4\sqrt{2}}{-2 - 2\sqrt{2} \left(\sqrt{\frac{7}{6}}x - \frac{iy}{\sqrt{6}} \right)^2} + \frac{1}{\sqrt{2}} \right).$$

The plot for real and imaginary part of this potential is shown in Fig. 4.

$\mathbf{m_1, m_2}$	$\mathbf{V_{RE,m_1,m_2}(x, y)}$
0, 0	$V(x, y) - \tilde{\omega}_1 - \tilde{\omega}_2$
0, 2	$V(x, y) - \tilde{\omega}_1 - \tilde{\omega}_2 + \frac{4\tilde{\omega}_2(\tilde{\omega}_2(ay+bx)^2-1)}{(\tilde{\omega}_2(ay+bx)^2+1)^2}$
2, 2	$V(x, y) - \tilde{\omega}_1 - \tilde{\omega}_2 + \frac{4\tilde{\omega}_2(\tilde{\omega}_2(ay+bx)^2-1)}{(\tilde{\omega}_2(ay+bx)^2+1)^2} + \frac{4\tilde{\omega}_1(\tilde{\omega}_1(ax-by)^2-1)}{((ax-by)^2\tilde{\omega}_1+1)^2}$
2, 4	$V(x, y) - \tilde{\omega}_1 - \tilde{\omega}_2$ $+ \frac{4(2-(b(x-2y)+a(2x+y))(a(-2x+y)+b(x+2y))\tilde{\omega}_1+(ax-by)^2(-2abxy+a^2(2x^2+y^2)+b^2(x^2+2y^2))\tilde{\omega}_1^2)}{(bx+ay)^2(1+(ax-by)^2\tilde{\omega}_1)^2}$ $+ \frac{4(42-18(bx+ay)^2\tilde{\omega}_2)}{(bx+ay)^2(3+(bx+ay)^2\tilde{\omega}_2(6+(bx+ay)^2\tilde{\omega}_2))} + \frac{576(-1-2(bx+ay)^2\tilde{\omega}_2)}{(3bx+3ay+6(bx+ay)^3\tilde{\omega}_2+(bx+ay)^5\tilde{\omega}_2^2)^2}$

Table 4: Potentials $V_{RE,m_1,m_2}(x, y)$ for even (m_1, m_2) values in old coordinates.



Figure 3: Plots of the rationally extended potentials $V_{RE,m_1,m_2}(x, y)$ as functions of x and y for different values of m_1 and m_2 .

Figure Missing

$$\text{Re}[V_{RE,m_1=0,m_2=2}(x, y)]$$

Figure Missing

$$\text{Re}[V_{RE,m_1=2,m_2=2}(x, y)]$$

Figure Missing

$$\text{Im}[V_{RE,m_1=0,m_2=2}(x, y)]$$

Figure Missing

$$\text{Im}[V_{RE,m_1=2,m_2=2}(x, y)]$$

Figure 4: Plots of $V_{RE,m_1,m_2}(x, y)$ vs. x and y , showing the Real part (left) and Imaginary part (right) for various values of m_1 and m_2 , with imaginary λ .

4 Three-dimensional QAAO potentials

We consider a three-dimensional QAAO potentials (defined on the full-line) with combination of a linear and a quadratic perturbations $\lambda_0 z + \frac{\lambda}{2} xy$ or of only quadratic perturbations $(\lambda_1 xy + \lambda_2 yz + \lambda_3 zx)$. In each case, we will decouple the perturbed potential using a coordinate transformations.

4.1 Combination of a linear and a quadratic perturbations

In this section, we consider a three-dimensional QAAO potential $V^{lq}(x, y, z)$ with a linear and a quadratic perturbations, given by

$$V^{lq}(x, y, z) = \frac{1}{4} (\omega_x^2 x^2 + \omega_y^2 y^2 + \omega_z^2 z^2) + \lambda_0 z + \frac{\lambda}{2} xy. \quad (39)$$

which can be written in terms of one-dimensional potential with linear perturbation $V(\omega_z, z, \lambda_0)$ and two-dimensional anisotropic potential with a quadratic perturbation $V(x, y)$ as

$$V^{lq}(x, y, z) = V(x, y) + V^l(z), \quad (40)$$

which can be further decoupled in the tilde co-ordinates using the transformations (3) and (19) as

$$\tilde{V}^{lq}(\tilde{x}, \tilde{y}, \tilde{z}) = \tilde{V}(\tilde{x}, \tilde{y}) + \tilde{V}^l(\tilde{z}). \quad (41)$$

Thus, the eigenfunctions of this equation can be easily obtained by using Eqs. (5) and (20) and given as

$$\tilde{\psi}_{n_1, n_2, n_3}^{lq}(\tilde{x}, \tilde{y}, \tilde{z}) \propto \tilde{\psi}_{n_1, n_2}(\tilde{x}, \tilde{y}) \tilde{\psi}_{n_3}^l(\tilde{z}) \quad (42)$$

and the energy eigenvalues are given by the sum of energy eigenvalues of 2D and 1D cases as

$$\tilde{E}_{n_1, n_2, n_3}^{lq}(\tilde{\omega}_1, \tilde{\omega}_2, \tilde{\omega}_3) = \tilde{E}_{n_1, n_2}(\tilde{\omega}_1, \tilde{\omega}_2) + \tilde{E}_{n_3}(\tilde{\omega}_3) \quad (43)$$

In the hermitian case, both parameters λ_0 and λ are real, and hence the energy spectrum will be real only when the equation (22) is satisfied. However, in the non-Hermitian and $\mathcal{PT}$ -symmetric cases, one or both of the parameters must be imaginary. Depending on the real and imaginary values of λ and λ_0 , the following cases of non-Hermitian and $\mathcal{PT}$ -symmetric systems arise:

1. When only λ_0 is imaginary (say $\lambda_0 = i\gamma_0$)

The system is $\mathcal{PT}$ -symmetric for the parity operator $\mathcal{P}_2$ given in Eq. (B.5) given by

$$\mathcal{P}_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}. \quad (44)$$

Here the $\mathcal{PT}$ -symmetry is unbroken and the energy eigenvalues (43) will be real for all values of γ_0 and the eigenfunctions (42) satisfy

$$\mathcal{P}_2 T \tilde{\psi}_{n_1, n_2, n_3}^{lq}(\tilde{x}, \tilde{y}, \tilde{z}) = (-1)^{n_3} \tilde{\psi}_{n_1, n_2, n_3}^{lq}(\tilde{x}, \tilde{y}, \tilde{z}).$$

2. When only λ is imaginary (say $\lambda = i\gamma$)

The potential is $\mathcal{PT}$ symmetric under the parity operators $\mathcal{P}_1$ and $\mathcal{P}_3$ given by

$$\mathcal{P}_1 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}; \quad \mathcal{P}_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (45)$$

respectively. Simialr to the 2D case, the spectrum will be real when the conditions (23) are satisfied and the eigenfunctions (42) satisfy

$$\mathcal{P}_1 T \tilde{\psi}_{n_1, n_2, n_3}^{lq}(\tilde{x}, \tilde{y}, \tilde{z}) = \mathcal{P}_3 T \tilde{\psi}_{n_1, n_2, n_3}^{lq}(\tilde{x}, \tilde{y}, \tilde{z}) = (-1)^{n_1+n_2} \tilde{\psi}_{n_1, n_2, n_3}^{lq}(\tilde{x}, \tilde{y}, \tilde{z}).$$

3. When both λ_0 and λ are imaginary

In this case, the potential is still $\mathcal{PT}$ symmetric under the parity operators $\mathcal{P}_4$ given by

$$\mathcal{P}_4 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \quad (46)$$

and the eigenfunctions satisfy

$$\mathcal{P}_4 T \tilde{\psi}_{n_1, n_2, n_3}^{lq}(\tilde{x}, \tilde{y}, \tilde{z}) = (-1)^{n_1+n_2+n_3} \tilde{\psi}_{n_1, n_2, n_3}^{lq}(\tilde{x}, \tilde{y}, \tilde{z}).$$

4.1.1 Rational Extension of $\tilde{V}^{lq}(\tilde{x}, \tilde{y}, \tilde{z})$

The rational extension of the real and all the $\mathcal{PT}$ -symmetric cases of the potential $\tilde{V}^{lq}(\tilde{x}, \tilde{y}, \tilde{z})$ in the tilde co-ordinates is simply the sum of the rational extensions of the 2D AHO and the 1D AHO potentials given by Eqs. (26) and (12) respectively as

$$\tilde{V}_{RE, m_1, m_2, m_3}^{lq}(\tilde{x}, \tilde{y}, \tilde{z}) = \tilde{V}_{RE, m_1, m_2}(\tilde{x}, \tilde{y}) + \tilde{V}_{RE, m_3}^l(\tilde{z}) \quad (47)$$

where $\tilde{V}_{RE, m_1, m_2}(\tilde{x}, \tilde{y})$ and $\tilde{V}_{RE, m_3}^l(\tilde{z})$ are rationally extended potentials corresponding to the potentials $\tilde{V}(\tilde{x}, \tilde{y})$ and $\tilde{V}^l(\tilde{z})$ given by Eqs. (19) and (3) respectively. To get the above extended potential in terms of old co-ordinates, we use the inverse co-ordinate trasformations. Thus, the ground and the excited state eigenfunctions with their corresponding energy eigenvalues are given by

$$\tilde{\psi}_{RE, m_1, m_2, m_3, 0, 0, 0}^{lq}(\tilde{x}, \tilde{y}, \tilde{z}) \propto \tilde{\psi}_{RE, m_1, m_2, 0, 0}(\tilde{x}, \tilde{y}) \tilde{\psi}_{RE, m_3, 0}^l(\tilde{z}), \quad (48)$$

$$\tilde{\psi}_{RE, m_1, m_2, m_3, n_1+1, n_2+1, n_3+1}^{lq}(\tilde{x}, \tilde{y}, \tilde{z}) \propto \tilde{\psi}_{RE, m_1, m_2, n_1+1, n_2+1}(\tilde{x}, \tilde{y}) \tilde{\psi}_{RE, m_3, n_3+1}^l(\tilde{z}) \quad (49)$$

and

$$\tilde{E}_{RE, m_1, m_2, m_3, n_1+1, n_2+1, n_3+1}^{lq}(\tilde{\omega}_1, \tilde{\omega}_2, \tilde{\omega}_3) = \tilde{E}_{RE, m_1, m_2, n_1+1, n_2+1}(\tilde{\omega}_1, \tilde{\omega}_2) + \tilde{E}_{RE, m_3, n_3+1}^l(\tilde{\omega}_3) \quad (50)$$

with

$$\tilde{E}_{RE, m_1, m_2, m_3, 0, 0, 0}^{lq}(\tilde{\omega}_1, \tilde{\omega}_2, \tilde{\omega}_3) = \tilde{E}_{RE, m_1, m_2, 0, 0}(\tilde{\omega}_1, \tilde{\omega}_2) + \tilde{E}_{RE, m_3, 0}^l(\tilde{\omega}_3)$$

respectively. Here $n_1, n_2, n_3 = 0, 1, 2, \dots$. Similar to the conventional cases, there are three possibilities to make the system complex and $\mathcal{PT}$ -symmetric. The spectrum of all these systems are also real and the $\mathcal{PT}$ -symmetry is unbroken under the restrictions defined in the Eqs. (22) and (23). However, the values of m_1, m_2 and m_3 are different for different cases discussed as below:

- **When both λ and λ_0 are real:** In this case, the potential will be regular for even values of m_1, m_2 and m_3 .
- **When λ is real and λ_0 is imaginary:** In this case, the potential will be regular for even values of m_1, m_2 and all values (even and odd) of m_3 .
- **When λ is imaginary and λ_0 is real:** In this case, m_1, m_2 and m_3 all are even.
- **When both λ and λ_0 are imaginary:** Here, m_1, m_2 are even and all values of m_3 (even and odd).

4.2 3D QAAO with Combinations of Quadratic Perturbations

Now, we consider another example of a three-dimensional QAAO potential with a combination of quadratic perturbations given by

$$V(x, y, z) = \frac{1}{4} (\omega_x^2 x^2 + \omega_y^2 y^2 + \omega_z^2 z^2 + 2\lambda_1 xy + 2\lambda_2 yz + 2\lambda_3 zx). \quad (51)$$

The perturbed potential can, in principle, be decoupled into the standard form of QHO by applying a coordinate transformation. However, it is important to note that for a general perturbation of the form $\lambda_1 xy + \lambda_2 yz + \lambda_3 zx$, it is not always possible to find a general coordinate transformation with arbitrary frequencies that will reduce the potential to a pure anisotropic oscillator in the new coordinates. Transformations can, however, be derived for specific cases. For example, when two frequencies are equal, such as $\omega_x = \omega_y = \omega$ (say), the potential in (51) simplifies to

$$V^q(x, z, y) = \frac{1}{4} (\omega^2(x^2 + y^2) + \omega_z^2 z^2 + 2\lambda_1 xy + 2\lambda_2 yz + 2\lambda_3 zx). \quad (52)$$

The general transformations reduce to two special cases: either $\lambda_1 = 0$ with $\lambda_2 \neq \lambda_3$, and $\lambda_1 \neq 0$ with $\lambda_2 = \lambda_3$. These cases will be analyzed in the following subsection under the assumption of positive λ values.

4.2.1 For $\lambda_1 = 0$ and $\lambda_2 \neq \lambda_3$

In this case, the potential (52) is defined as

$$V^{q1}(x, y, z) = \frac{1}{4} (\omega^2(x^2 + y^2) + \omega_z^2 z^2 + 2\lambda_2 yz + 2\lambda_3 zx) \quad (53)$$

To solve this problem, we use the following coordinate transformations from old to new and vice-versa given as

$$\begin{aligned} x &= ad\tilde{y} + bd\tilde{z} - c\tilde{x}, \\ y &= d\tilde{x} + ac\tilde{y} + bc\tilde{z}, \\ z &= a\tilde{z} - b\tilde{y}, \end{aligned} \quad \begin{cases} \tilde{x} = -cx + d y, \\ \tilde{y} = ad x + acy - bz, \\ \tilde{z} = bd x + bcy + az. \end{cases} \quad (54)$$

where $k = \frac{\omega^2 - \omega_z^2}{\sqrt{4\lambda_2^2 + 4\lambda_3^2 + (\omega^2 - \omega_z^2)^2}}$, $c = \frac{\lambda_2}{\sqrt{\lambda_2^2 + \lambda_3^2}}$ and $d = \frac{\lambda_3}{\sqrt{\lambda_2^2 + \lambda_3^2}}$. The other two parameters a and b are same as defined in the 2D case by Eq. (17). Thus, the potential in the new co-ordinate will be

$$\tilde{V}^q(\tilde{x}, \tilde{y}, \tilde{z}) = \frac{1}{4} (\tilde{\omega}_1^2 \tilde{x}^2 + \tilde{\omega}_2^2 \tilde{y}^2 + \tilde{\omega}_3^2 \tilde{z}^2) \quad (55)$$

where the new frequencies are given by

$$\begin{aligned}\tilde{\omega}_1 &= \omega, \\ \tilde{\omega}_2 &= \sqrt{\frac{1}{2} \left(\omega^2 + \omega_3^2 - \sqrt{4(\lambda_2^2 + \lambda_3^2) + (\omega^2 - \omega_3^2)^2} \right)}, \\ \tilde{\omega}_3 &= \sqrt{\frac{1}{2} \left(\omega^2 + \omega_3^2 + \sqrt{4(\lambda_2^2 + \lambda_3^2) + (\omega^2 - \omega_3^2)^2} \right)}.\end{aligned}\tag{56}$$

Rational Extension:

Similar to the 2D case, to get the rational extension of the above potential (53), we use the starting potential (55) and get the corresponding RE potential in the new co-ordinates which is transformed back to the old co-ordinates with perturbation terms using the inverse co-ordinate transformation (54). The expression of the RE potential in the new co-ordinate is given as

$$\tilde{V}_{RE,m_1,m_2,m_3}^{q_1}(\tilde{x}, \tilde{y}, \tilde{z}) = \tilde{V}_{RE,m_1}(\tilde{x}) + \tilde{V}_{RE,m_2}(\tilde{y}) + \tilde{V}_{RE,m_3}(\tilde{z})\tag{57}$$

where $(\tilde{V}_{RE,m_1}(\tilde{x}), \tilde{V}_{RE,m_2}(\tilde{y}), \tilde{V}_{RE,m_3}(\tilde{z}))$ are given by Eq. (7) but in new angular frequency and coordinates. Similarly the ground and the excited state eigenfunctions are written using rationally extended eigenfunctions in new coordinates and angular frequency as

$$\tilde{\psi}_{RE,m_1,m_2,m_3,0,0,0}^{q_1}(\tilde{x}, \tilde{y}, \tilde{z}) \propto \tilde{\psi}_{RE,m_1,0}(\tilde{x}) \tilde{\psi}_{RE,m_2,0}(\tilde{y}) \tilde{\psi}_{RE,m_3,0}(\tilde{z})$$

and

$$\tilde{\psi}_{RE,m_1,m_2,m_3,n_1+1,n_2+1,n_3+1}^{q_1}(\tilde{x}, \tilde{y}, \tilde{z}) \propto \tilde{\psi}_{RE,m_1,n_1+1}(\tilde{x}) \tilde{\psi}_{RE,m_2,n_2+1}(\tilde{y}) \tilde{\psi}_{RE,m_3,n_3+1}(\tilde{z})\tag{58}$$

respectively. The corresponding energy eigenvalues in new angular frequency are given as

$$\begin{aligned}\tilde{E}_{RE,m_1,m_2,m_3,n_1+1,n_2+1,n_3+1}^{q_1}(\tilde{\omega}_1, \tilde{\omega}_2, \tilde{\omega}_3) &= \tilde{E}_{RE,m_1,n_1+1}(\tilde{\omega}_1) \\ &+ \tilde{E}_{RE,m_2,n_2+1}(\tilde{\omega}_2) + \tilde{E}_{RE,m_3,n_3+1}(\tilde{\omega}_3)\end{aligned}\tag{59}$$

with

$$\tilde{E}_{RE,m_1,m_2,m_3,0,0,0}^{q_1}(\tilde{\omega}_1, \tilde{\omega}_2, \tilde{\omega}_3) = 0\tag{60}$$

The conditions on λ 's for real spectra are as follows:

1. **Hermitian case:** When λ 's are real, then real spectra are seen under the following condition

$$0 \leq \lambda_2^2 + \lambda_3^2 \leq \omega^2 \omega_3^2\tag{61}$$

2. **Non-Hermitian and $\mathcal{PT}$ -symmetric case:** With imaginary λ 's, the system shows η -pseudo-hermiticity and hence the system is $\mathcal{PT}$ -symmetric for the different parity operators $\mathcal{P}$ depending on whether one or both the coupling parameters (λ_2 and λ_3) are imaginary. Thus, we have three different cases discussed as below:

(a) **When λ_3 is real and λ_2 is imaginary (say $\lambda_2 = i\gamma_2$)**

The system exhibits $\mathcal{PT}$ symmetry with the parity operator $\mathcal{P}_3$ (Eq. (B.5)), defined as

$$\mathcal{P}_3 : x \rightarrow x, \quad y \rightarrow -y, \quad z \rightarrow z.$$

and the real spectra is seen when

$$-\frac{1}{4}(\omega^2 - \omega_3^2)^2 \leq -\gamma_2^2 + \lambda_3^2 \leq \omega^2 \omega_3^2.$$

(b) When λ_3 is imaginary (say $\lambda_3 = i\gamma_3$) and λ_2 is real

Here $\mathcal{PT}$ symmetry is defined by the parity operator $\mathcal{P}_1$ (Eq. (B.5)), where

$$\mathcal{P}_1 : x \rightarrow -x, \quad y \rightarrow y, \quad z \rightarrow z.$$

and the spectrum will be real when

$$-\frac{1}{4}(\omega^2 - \omega_3^2)^2 \leq \lambda_2^2 - \gamma_3^2 \leq \omega^2 \omega_3^2.$$

(c) When both λ 's are imaginary (say, $\lambda_2 = i\gamma_2$ and $\lambda_3 = i\gamma_3$)

In this case the system is $\mathcal{PT}$ symmetric for the parity operator $\mathcal{P}_2$ (Eq. (B.5))

$$\mathcal{P}_2 : x \rightarrow x, \quad y \rightarrow y, \quad z \rightarrow -z,$$

and the real spectra are observed under the following conditions

$$\gamma_2^2 + \gamma_3^2 \leq \frac{1}{4}(\omega^2 - \omega_3^2)^2.$$

Conditions for Degeneracy:

The condition for degeneracy in the system is given similar to the 2D case except λ is replaced by $\sqrt{\lambda_2^2 + \lambda_3^2}$ and is given by

$$\sqrt{\lambda_2^2 + \lambda_3^2} = \frac{\sqrt{(\tilde{u}^4 + \tilde{v}^4) - \left(\frac{\tilde{u}\tilde{v}}{\tilde{u}\tilde{v}}\right)^2 (u^4 + v^4)}}{\tilde{u}^2 + \tilde{v}^2} \omega \omega_3 \quad (62)$$

where $\tilde{\omega}_2 : \tilde{\omega}_3 = \tilde{u} : \tilde{v}$ is a rational ratio and $\omega : \omega_3 = u : v$ can be rational or irrational. As an illustration, we consider the case of three unequal frequencies (i.e., $\omega : \omega_z = \sqrt{2} : 1$).

$$V(x, y, z) = 2x^2 + 2y^2 + z^2 + \frac{2}{5}\sqrt{7}yz + \frac{2}{5}\sqrt{7}xz$$

In this case, the coordinate transformation can be carried out by substituting the parameter k to $\frac{5}{9}$ as

$$\begin{aligned} x &= -\frac{\tilde{x}}{\sqrt{2}} + \frac{\tilde{y}}{3} + \frac{1}{3}\sqrt{\frac{7}{2}}\tilde{z} \\ y &= \frac{\tilde{x}}{\sqrt{2}} + \frac{\tilde{y}}{3} + \frac{1}{3}\sqrt{\frac{7}{2}}\tilde{z} \\ z &= -\frac{\sqrt{7}\tilde{y}}{3} + \frac{\sqrt{2}\tilde{z}}{3} \end{aligned}$$

The potential in the new coordinates simplifies to

$$\tilde{V}(\tilde{x}, \tilde{y}, \tilde{z}) = 2\tilde{x}^2 + \frac{3}{5}\tilde{y}^2 + \frac{12}{5}\tilde{z}^2$$

The degeneracy condition simplifies to

$$\sqrt{\lambda_2^2 + \lambda_3^2} = \frac{\sqrt{2(\tilde{u}^2 - \tilde{v}^2)^2 - (\tilde{u}\tilde{v})^2}}{\tilde{u}^2 + \tilde{v}^2} \omega^2 \quad (63)$$

For example, the following values of (λ_2, λ_3) , in units of ω^2 , result in $\tilde{u} : \tilde{v} = 2 : 1$ and therefore lead to degeneracy:

$$(\lambda_2, \lambda_3) = \left(\frac{\sqrt{7}}{5}, \frac{\sqrt{7}}{5} \right), \left(\frac{\sqrt{6}}{5}, \frac{\sqrt{8}}{5} \right), \left(0, \frac{\sqrt{14}}{5} \right), \dots$$

4.2.2 For $\lambda_1 \neq 0$ and $|\lambda_2| = |\lambda_3|$

Here, if we consider $\lambda_2 = \lambda_3 = \lambda$ (say), with λ being real or imaginary, the potential (51) becomes

$$V^{q_2}(x, y, z) = \frac{1}{4} (\omega^2(x^2 + y^2) + \omega_z^2 z^2 + 2\lambda_1 xy + 2\lambda(yz + zx)). \quad (64)$$

The transformation relations from old to new coordinates and vice-versa are given by,

$$\begin{cases} x = \frac{a}{\sqrt{2}}\tilde{y} + \frac{b}{\sqrt{2}}\tilde{z} - \frac{\tilde{x}}{\sqrt{2}}, \\ y = \frac{a}{\sqrt{2}}\tilde{y} + \frac{b}{\sqrt{2}}\tilde{z} + \frac{\tilde{x}}{\sqrt{2}}, \\ z = a\tilde{z} - b\tilde{y}, \end{cases} \begin{cases} \tilde{x} = \frac{y}{\sqrt{2}} - \frac{x}{\sqrt{2}}, \\ \tilde{y} = \frac{a}{\sqrt{2}}x + \frac{a}{\sqrt{2}}y - bz, \\ \tilde{z} = \frac{b}{\sqrt{2}}x + \frac{b}{\sqrt{2}}y + az. \end{cases} \quad (65)$$

where $k = \frac{\omega^2 - \omega_3^2 + q\lambda_1}{\sqrt{8\lambda^2 + (\omega^2 - \omega_3^2 + q\lambda_1)^2}}$, and the new frequencies are

$$\begin{aligned} \tilde{\omega}_1 &= \sqrt{\omega^2 - \lambda_1}, \\ \tilde{\omega}_2 &= \sqrt{\frac{1}{2} \left(\omega^2 + \omega_3^2 + \lambda_1 - \sqrt{8\lambda^2 + (\omega^2 - \omega_3^2 + \lambda_1)^2} \right)}, \\ \tilde{\omega}_3 &= \sqrt{\frac{1}{2} \left(\omega^2 + \omega_3^2 + \lambda_1 + \sqrt{8\lambda^2 + (\omega^2 - \omega_3^2 + \lambda_1)^2} \right)}. \end{aligned} \quad (66)$$

Rational Extension

The rational extension of this potential is similar to the previous case except the new frequencies and new coordinates are given by Eqs. (66) and (65) respectively.

The condition on λ 's for real spectra: From the expressions for the new angular frequencies (66), it is to be noted that the parameter λ_1 is always real, whereas the parameter λ can be either real or imaginary. This distinction gives rise to two possibilities for real spectra and the conditions for real spectra in each of these cases are discussed below:

1. **Hermitian case:** When λ is real, then the real spectra are observed under the following conditions on λ_1 and λ given as

$$\begin{aligned} -\omega^2 &\leq \lambda_1 \leq \omega^2 \\ \text{and } 0 &\leq \lambda_1^2 \leq \frac{(\omega^2 + \lambda)\omega_z^2}{2}. \end{aligned} \quad (67)$$

2. **Non-Hermitian and $\mathcal{PT}$ symmetric case:** When λ is imaginary, say $\lambda = i\gamma$, the allowed values of γ^2 for various ranges of λ_1 and ω are

$$-\omega^2 \leq \lambda_1 \leq \omega^2 \quad \text{and} \quad 0 \leq \gamma^2 \leq \frac{(\omega^2 - \omega_3^2 + \lambda_1)^2}{8} \quad (68)$$

Since the system shows η -pseudo-hermiticity and hence the system has $\mathcal{PT}$ -symmetry for some parity operator $\mathcal{P}_2$ (Eq. (B.5)) i.e.,

$$\mathcal{P}_2 : x \rightarrow x, \quad y \rightarrow y, \quad z \rightarrow -z.$$

The above wavefunction when expressed in terms of old coordinates satisfy following relations

$$\mathcal{PT} \tilde{\psi}_{n_1, n_2, n_3}(\tilde{x}, \tilde{y}, \tilde{z}) = \pm \tilde{\psi}_{n_1, n_2, n_3}(\tilde{x}, \tilde{y}, \tilde{z})$$

as the new coordinates satisfy $\mathcal{PT}$ symmetry when expressed in terms of old coordinates

$$\mathcal{PT} \tilde{x}(x, z, y) = \pm \tilde{x}(x, z, y)$$

$$\mathcal{PT} \tilde{y}(x, z, y) = \pm \tilde{y}(x, z, y)$$

$$\mathcal{PT} \tilde{z}(x, z, y) = \pm \tilde{z}(x, z, y)$$

Degeneracy Conditions:

Similar to the previous case, the condition for degeneracy in this case is given by

$$\sqrt{8\lambda^2 + \lambda_1^2} = \frac{|\tilde{v}^2 - \tilde{u}^2|}{\tilde{v}^2 + \tilde{u}^2} (2\omega^2 + \lambda_1), \quad (69)$$

which simplifies to following value of λ as

$$\lambda = \frac{\sqrt{\left(\frac{|\tilde{v}^2 - \tilde{u}^2|}{\tilde{v}^2 + \tilde{u}^2} (\lambda_1 + 2\omega^2)\right)^2 - \lambda_1^2}}{2\sqrt{2}}. \quad (70)$$

Whenever this condition is satisfied, for some integer values of $\tilde{u}, \tilde{v}$, the system will be degenerate. For example, when $\lambda = \frac{\omega^2}{4} \sqrt{\frac{15}{2}}$ and $\lambda_1 = \frac{\omega^2}{2}$, resulting in $\tilde{u} : \tilde{v} = 1 : 3$, the system exhibits degeneracy.

5 Summary and Discussion

In this work, we explored the rational extensions of the quantum anisotropic harmonic oscillator (QAHO) under various perturbations in higher dimensions with their exact bound state solutions. First, we consider a simple case of 1D QHO with a linear perturbation and shown that the rational extensions exist for all integers of m when the perturbation is imaginary but only of even m when perturbation is real. It is also shown that the energy eigenvalues of the RE potentials are isospectral to the conventional one and the spectrum is always real in both the Hermitian as well as non-Hermitian $\mathcal{PT}$ -symmetric cases. The expressions for the potentials with imaginary perturbations and the corresponding ground states for different m are derived and shown their behaviors graphically.

In the higher dimensional cases with quadratic perturbations, we decouple the potentials via similarity transformations, reducing them to standard QAHO forms. Systems with imaginary perturbations exhibit $\mathcal{PT}$ -symmetry, supported by suitable parity operators. The conditions for the real spectra (unbroken $\mathcal{PT}$ -symmetry) with degeneracy conditions are also discussed. As an example, we consider 2D-QAHO potential with quadratic perturbations and obtained the corresponding RE potentials with their exact solutions for even co-dimensions of m_1 and m_2 . The conditions for the real spectra and the degeneracy in both the Hermitian and the non-Hermitian $\mathcal{PT}$ -symmetric cases are discussed. The behaviors of these potentials (real and complex) with their ground state solutions are also shown graphically for different combinations of m_1 and m_2 .

As a last example, we consider the case of 3D QAHO potentials with combined linear and/or quadratic perturbations. First, we consider the potentials with linear and quadratic perturbations and constructed a three parameter (m_1, m_2, m_3) family of the RE Hermitian as well as non-Hermitian $\mathcal{PT}$ -symmetric systems with their exact bound state solutions. Conditions on perturbation parameters and angular frequencies ensuring real spectra for all possible combinations of m_1, m_2 and m_3 are also

analyzed. Thereafter, we consider a more general form of the potential with combinations of quadratic perturbations only and solved this problem for some specific restrictions on perturbations parameters. The rational extension corresponding to these potentials with the degeneracy conditions are also discussed.

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Appendix A: Similarity Transformation

The Schrödinger equation for a quantum system with a Hamiltonian H having coupled potential (potential with some perturbation term real or imaginary) is given by

$$\hat{H} |\psi_n\rangle = E_n |\psi_n\rangle, \quad (\text{A.1})$$

where $\hat{H}$ is the Hamiltonian operator, $|\psi_n\rangle$ is the eigenfunction, and E_n is the energy eigenvalue. If $\hat{H}$ is pseudo-hermitian [4], then there exists a Hamiltonian $\hat{H}_0$ isospectral to $\hat{H}$ in which the potential is decoupled into a set of exactly solvable independent potentials, and eigenvalues are real or occurs in complex conjugate pairs. If the eigenvalues of $\hat{H}$ are real, then the Hamiltonian $\hat{H}_0$ is said to be hermitian, and the basis set $\{|\tilde{\psi}_n\rangle\}$ is orthonormal. The Hamiltonian $\hat{H}$ can be transformed into $\hat{H}_0$ by a similarity transformation using an invertible linear operator U as

$$\hat{H}_0 = U \hat{H} U^{-1}, \quad (\text{A.2})$$

For a hermitian operator $\hat{H}$, the unitary operator U satisfies $U^{-1} = U^\dagger$. However, for non-hermitian operators $\hat{H}$, this relation does not hold. The eigenfunctions of $\hat{H}_0$ are connected to those of $\hat{H}$ by the relation

$$|\psi_n\rangle = U |\tilde{\psi}_n\rangle. \quad (\text{A.3})$$

For real spectra the Hamiltonian $\hat{H}_0$ is hermitian, and therefore $\hat{H}_0 = \hat{H}_0^\dagger$, and (A.2) can be rewritten as

$$\hat{H}_0 = (U^{-1})^\dagger \hat{H} U^\dagger. \quad (\text{A.4})$$

The eigenfunction $|\Psi_n\rangle$ of the adjoint operator $\hat{H}^\dagger$ is, therefore, given by

$$|\Psi_n\rangle = (U^{-1})^\dagger |\tilde{\psi}_n\rangle. \quad (\text{A.5})$$

Both sets of eigenfunctions $\{|\psi_n\rangle\}$ and $\{|\Psi_n\rangle\}$ form a biorthonormal basis. The basis set $\{|\psi_n\rangle\}$ is orthonormal with the metric $U^\dagger U = \eta$ (say) as

$$\langle \psi_n | \psi_m \rangle = \langle \tilde{\psi}_n | U^\dagger U | \tilde{\psi}_m \rangle = \langle \tilde{\psi}_n | \eta | \tilde{\psi}_m \rangle = \delta_{nm}. \quad (\text{A.6})$$

On the other hand, the standard inner product

$$\langle \psi_n | \psi_m \rangle = \langle \tilde{\psi}_n | (U^{-1})^\dagger U^{-1} | \tilde{\psi}_m \rangle = \delta_{nm}, \quad (\text{A.7})$$

is not necessarily finite. From (A.2) and (A.4), one can express the η -pseudo-hermiticity of the Hamiltonian $\hat{H}$ as

$$\hat{H}^\dagger = \eta \hat{H} \eta^{-1}. \quad (\text{A.8})$$

The commutation relation of operators $[A, B]_\psi$ and $[A', B']_{\tilde{\psi}}$ in the old and new basis, respectively, is given by

$$[A, B]_\psi = U^{-1} [A', B']_{\tilde{\psi}} U. \quad (\text{A.9})$$

In particular, the commutator $[x', p'] = iI$ between the coordinates x' and p' is conserved under the similarity transformation, and therefore the commutation relation in the coupled coordinate system is

$$[x, p] = iI. \quad (\text{A.10})$$

Appendix B: The concept of $\mathcal{PT}$ -Symmetry in higher dimensions

Here we will show that Dirac hermiticity can be generalized to non-hermitian Hamiltonians, which guarantees the reality of the spectra by introducing the concept of $\mathcal{PT}$ symmetry [1, 23]. Here $\mathcal{P}$ stands for parity denoting space symmetry, and $\mathcal{T}$ stands for time reversal symmetry, which is anti-linear. A Hamiltonian is said to be $\mathcal{PT}$ symmetric if it commutes with the product of the parity and time reversal operators, i.e.,

$$[H, \mathcal{PT}] = 0 \quad \text{and} \quad \mathcal{PT} |\psi\rangle = \pm |\psi\rangle, \quad (\text{B.1})$$

and the $\mathcal{PT}$ -symmetric Hamiltonian, $H^{\mathcal{PT}}$, preserves the physical axioms of quantum mechanics and satisfies the equation

$$\hat{H} = \chi \hat{H} \chi^{-1}, \quad (\text{B.2})$$

where χ denotes the $\mathcal{PT}$ operator, differing from η -pseudo-hermiticity in (A.8) as the similarity transform yields $\hat{H}$, not its adjoint. If the Hamiltonian commutes with the parity operator, i.e., $[H, \mathcal{P}] = 0$, then the eigenfunctions of the Hamiltonian are also the eigenfunctions of the parity operator, and the Hamiltonian is said to be parity symmetric. The operator $\mathcal{PT}$ is not linear like $\mathcal{P}$ because $\mathcal{T}$ is anti-linear, and therefore $\mathcal{PT}$ symmetry is more subtle than $\mathcal{P}$ symmetry. Due to the antilinear nature of $\mathcal{PT}$, the eigenstates of H may or may not be the eigenstates of $\mathcal{PT}$. There are two possibilities: either the eigenstates of H are also the eigenstates of $\mathcal{PT}$ or the eigenstates of H are not the eigenstates of $\mathcal{PT}$. The $\mathcal{PT}$ symmetry is *unbroken* if all the eigenstates of the Hamiltonian are also the eigenstates of the $\mathcal{PT}$ operator, and the system exhibits real spectra. The $\mathcal{PT}$ symmetry is said to be *broken* if the eigenstates of the Hamiltonian are not the eigenstates of the $\mathcal{PT}$ operator, and the system exhibits complex spectra. The parity ($\mathcal{P}$) and time ($\mathcal{T}$) transformations in 1-D are defined as

$$\begin{aligned} \mathcal{P} : x &\rightarrow -x, \quad i \rightarrow i \\ \mathcal{T} : x &\rightarrow x, \quad i \rightarrow -i \end{aligned}$$

In 2-D, the parity transform is defined in two discrete steps [12], say $\mathcal{P}_1$ and $\mathcal{P}_2$, as

$$\begin{aligned} \mathcal{P}_1 : x &= -x, \quad y = y \\ \mathcal{P}_2 : x &= x, \quad y = -y \end{aligned}$$

A matrix representation of $\mathcal{P}$ (in any number of dimensions) has a determinant equal to -1 , distinguishing it from a rotation, which has a determinant equal to $+1$. In matrix form, the above two parity operators are given by

$$\mathcal{P}_1 = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}; \quad \text{and} \quad \mathcal{P}_2 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad (\text{B.3})$$

respectively. There are other possible transformations, provided they all satisfy the condition that the determinant of the transformation matrix is -1 . This suggests that the parity operator is not unique and can be defined in many ways. Therefore, two more possibilities are

$$\mathcal{P}_3 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}; \quad \text{and} \quad \mathcal{P}_4 = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} \quad (\text{B.4})$$

Similarly, in 3D case, some parity transformation matrices are

$$\begin{aligned} \mathcal{P}_1 &= \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}; & \mathcal{P}_2 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}; \\ \mathcal{P}_3 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}; & \mathcal{P}_4 &= \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}. \end{aligned} \tag{B.5}$$

It can be seen that the first operator is defining the reflection about the plane $x = 0$, the second operator is defining the reflection about the plane $z = 0$, the third operator is defining the reflection about the plane $y = 0$, and the fourth operator is defining the space inversion.

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